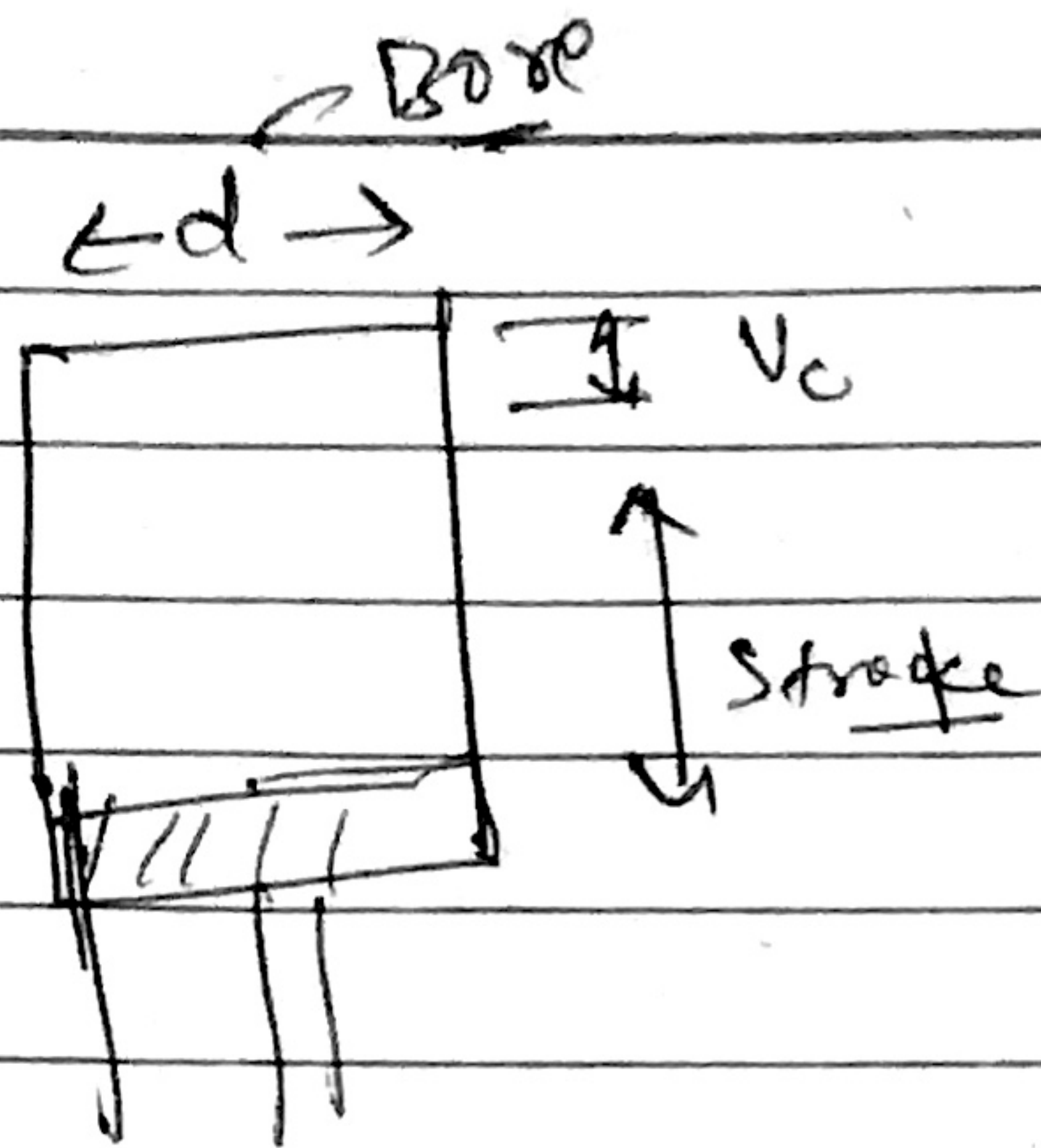


Engines

best ball

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$V_c \leftarrow$ Clearance.

$$V_s = \frac{\pi}{4} d^2 \text{ (Swept volume)}$$

Cubic capacity = $k \times V_s$

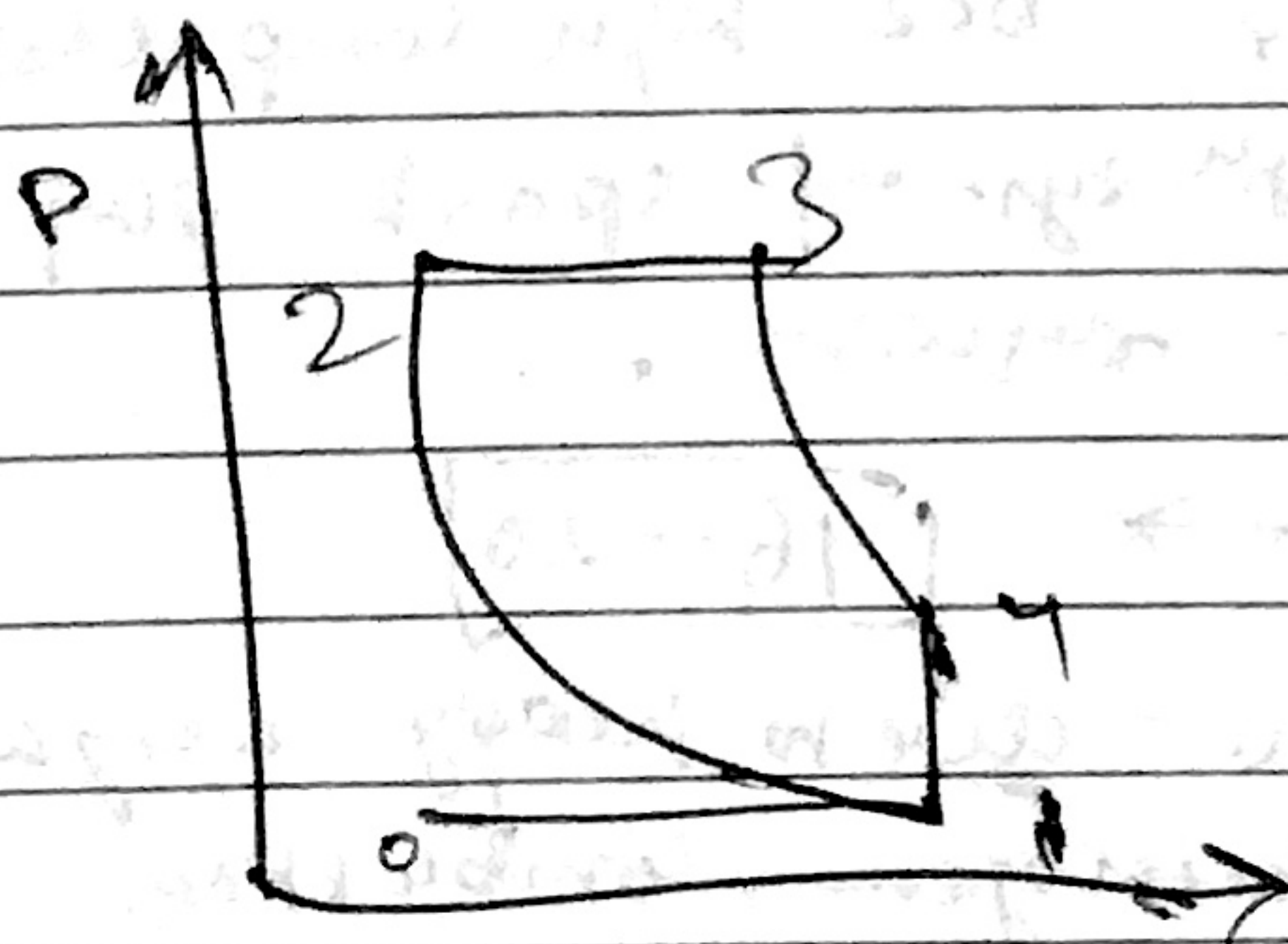
$k \leftarrow$ No. of cylinders

Compression Ratio $\rightarrow \gamma = \frac{V_s + V_c}{V_c}$

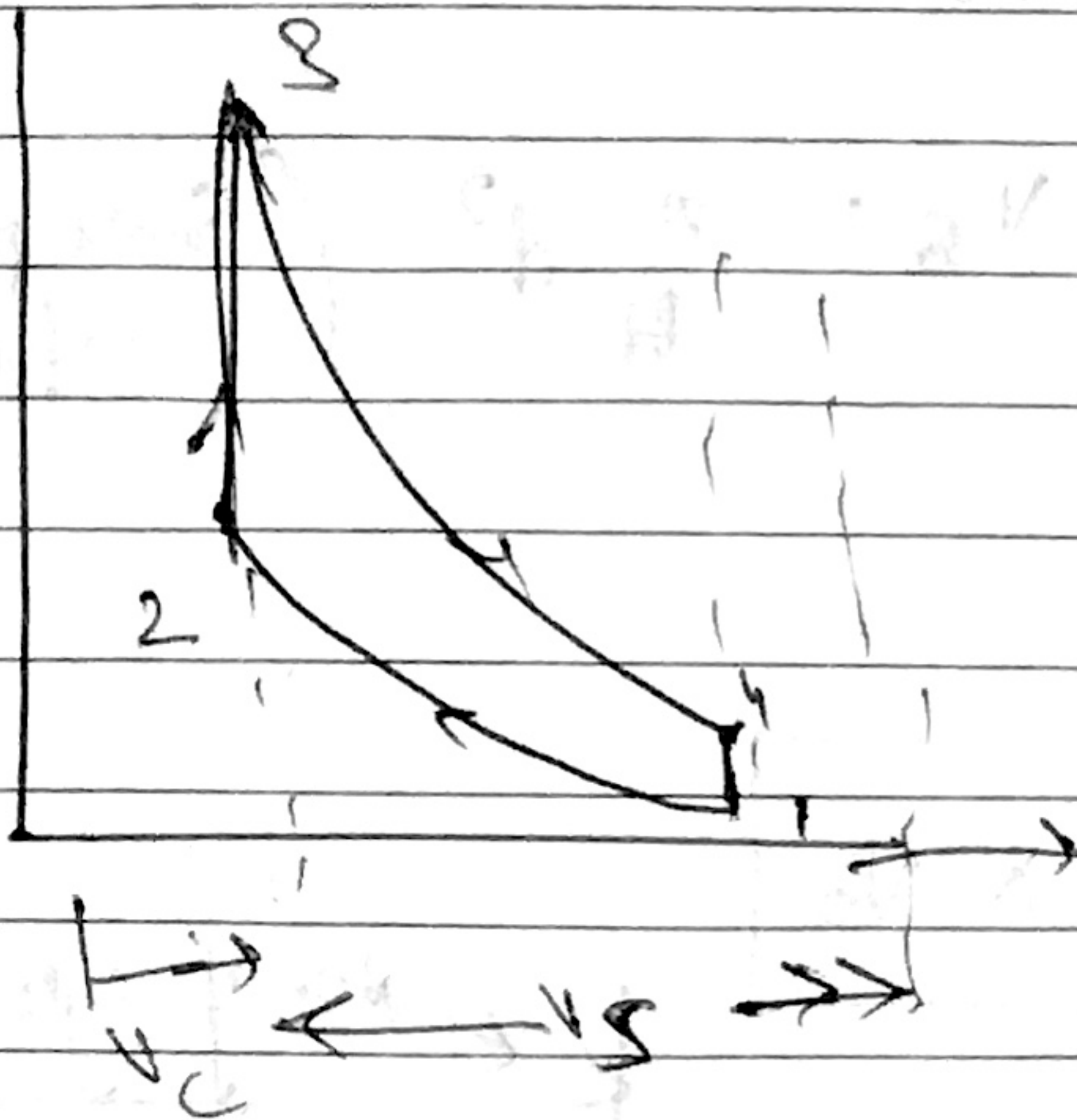
4-Stroke

Suction $\rightarrow$ Compression $\rightarrow$ Expansion

$\downarrow$
Exhaust



2-stroke



SI

CI

- | | |
|---|---|
| ① Otto cycle ($\Delta v = 0$) Q _{add} | ① Diesel ($\Delta P = 0$) Q _{add} |
| ② petrol/gasoline
high volatile | ② Diesel, non-volatile |
| ③ Gaseous mix. of air + fuel | ③ Direct fuel. at high pressure |
| ④ Carburetor + ignition system neces. | ④ Fuel pump + injector necessary |
| ⑤ Throttle control air-fuel
mix. (quantity) | ⑤ fuel quantity is regulated
air quantity is not. |
| ⑥ Req. igniti ⁿ system with spark
plug in combustion chamber
Primary voltage is provided by
Battery | ⑥ Self igniti ⁿ due to high temp.
of air bcz high compression.
→ igniti ⁿ sys. w/ spark plug
is not require. |
| ⑦ $r \rightarrow$ 6-10 | ⑦ $r \rightarrow$ 16-20 |
| ⑧ faster due to light weight
and homogeneous combustion | ⑧ Slow - due to heavy weight
and heterogeneous combustion. |
| ⑨ η - low - due to low CR | ⑨ η - high - due to high CR. |
| ⑩ lighter due to lower
peak pressure | ⑩ cleaner due to higher
peak pressure. |

Good Write

4-Stroke

- ① Cycle is completed in four strokes or 2-revolutions of crankshaft.
- ② 1 power stroke is obtained in 2-revolutions of crankshaft.
- ③ Heavier flywheel req.
- ④ lesser lubrication and cooling required.
- ⑤ Lower rate of wear and tear.
- ⑥ Valves control the entry of fuel and exit of exhaust.
- ⑦ Initial cost is high because of the apparatus (valves -).
- ⑧ Volumetric eff. is high due to more time in induction.
- ⑨ η is higher.
- ⑩ Used where eff. is imp. e.g. car, buses, industrial engines, planes, etc.

Knocking

2-Stroke

- ① Thermodynamic cycle is completed in 2-revolutions.
- ② 1 power stroke obtained in each revolution of crankshaft.
- ③ Lighter flywheel can be used.
- ④ High lubrication and cooling is required.
- ⑤ Higher rate of wear and tear.
- ⑥ No-valves, only ports.
- ⑦ Initial cost is low (might not have valves -).
- ⑧ Volumetric eff. is low.
- ⑨ η is lower.
- ⑩ Used where cost, compactness, and light weight are imp. e.g. mopeds, scooters, bikes, hand sprayers, etc.

Detonation